

Project #5: DESIGN AND DEVELOPMENT OF A MODULAR SOFT-RIGID DEXTEROUS ROBOTIC HAND

A sensor-rich, high-degree-of-freedom end-effector for intelligent manipulation

1. Project Description

This project proposes the design, implementation and experimental evaluation of a five-fingered soft-rigid dexterous robotic hand for industrial arms, collaborative robots, mobile manipulators and humanoid platforms. Rigid links and articulated joints will provide strength and accurate motion, while compliant silicone skin, replaceable fingertip pads and embedded tactile sensors will improve conformity, friction and safe contact with delicate or irregular objects.

The prototype will include an opposable thumb and at least 10 controllable degrees of freedom. The team will determine the minimum useful DoF arrangement for power, pinch, tripod, lateral and precision grasps, compare tendon-driven, direct-drive and hybrid actuation, and integrate perception, force/slip feedback, embedded control and a quick-mount wrist interface.

2. Why It Is Important

- Dexterity closes the gap between reaching an object and manipulating it reliably.
- Tactile feedback enables millisecond-level contact, force and slip correction.
- Soft-rigid construction combines strong power grasps with damage-free delicate handling.
- A modular platform can replace several task-specific grippers across manufacturing, agriculture and service robotics.

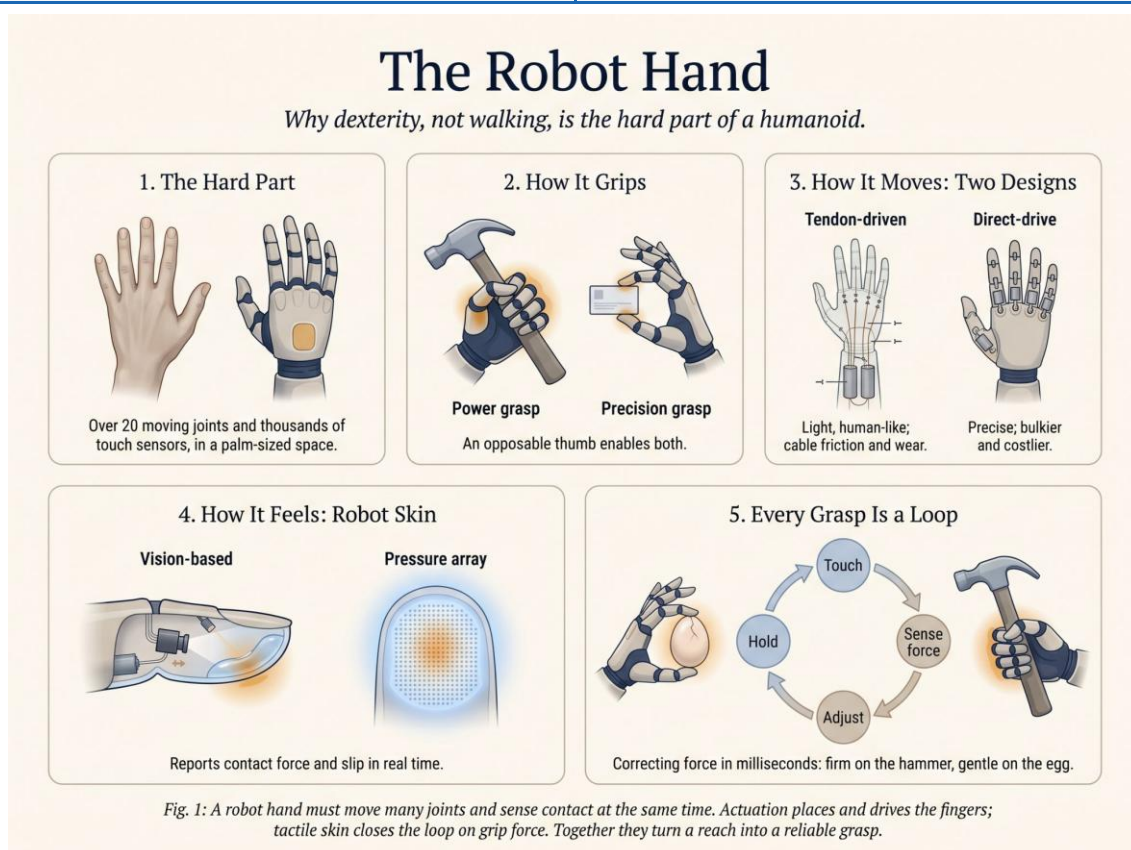


Figure 1. Dexterity, actuation alternatives, robot skin and the closed-loop nature of a reliable grasp.

3. Proposed System Layout and Main Components

The proposed system is organised as a modular mechatronic platform. The hand contains the soft-rigid fingers, palm structure, tendon or local-drive mechanisms, embedded sensing and motor electronics. A wrist adapter supplies power and communication and mechanically connects the hand to a host robot. Sensor provides object pose and grasp candidates, while tactile and proprioceptive feedback closes the control loop after contact.

SOFT-RIGID DEXTEROUS ROBOTIC HAND

A Modular, Sensor-Rich Hand for Intelligent Manipulation

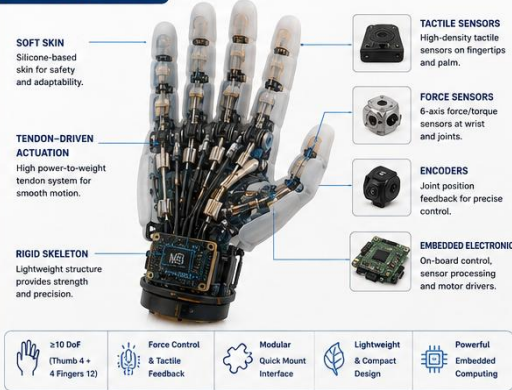
- High Dexterity**
≥10 DoF with human-like kinematics and motion.
- Soft-Rigid Hybrid Design**
Rigid skeleton for strength, soft skin for compliance and safe interaction.
- Rich Tactile Feedback**
Embedded force/pressure sensors for precise grip and slip detection.
- Modular & Versatile**
Easily mounts on robotic, arms, humanoid, or mobile platforms.
- Advanced Perception**
Tactile + proprioception + vision for robust, adaptive manipulation.



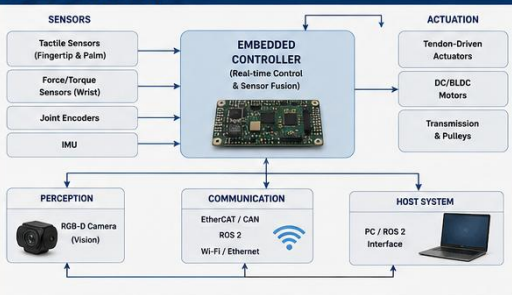
- SOFT EXTERIOR**
Conformable skin for safe contact and better grip.
- RIGID INTERNAL STRUCTURE**
Strong and lightweight mechanism for high payload and durability.



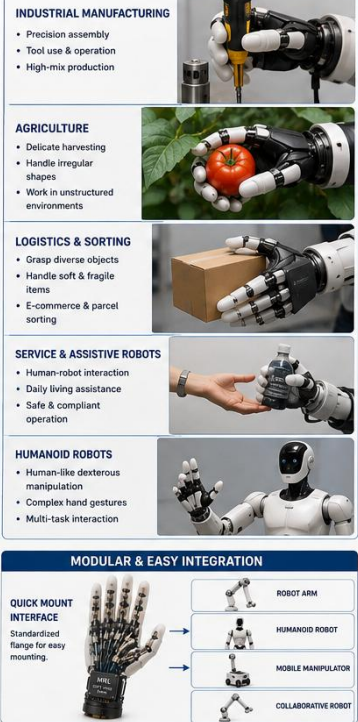
SYSTEM OVERVIEW



SYSTEM ARCHITECTURE



APPLICATIONS



- | KEY FEATURES | PERFORMANCE TARGETS | EVALUATION & TESTING | PROJECT OUTCOMES |
|--|---|---|---|
| <ul style="list-style-type: none"> ≥10 DoF for high dexterity Soft-rigid hybrid for safety and performance Embedded tactile & force sensing Modular design for wide compatibility Real-time control & advanced perception | <ul style="list-style-type: none"> Payload: ≥ 2 kg Grip Force: ≥ 25 N per finger Tactile Resolution: < 1 mm Response Time: < 50 ms Weight: < 1.5 kg | <ul style="list-style-type: none"> Standard Grasping Tests (ASTM F2992) Dexterity Tests (Johansen-Taylor) Tool Use Benchmarks Tactile Sensing Accuracy Real-world Application Trials | <ul style="list-style-type: none"> A high-performance, modular dexterous hand Open-source software & hardware interfaces Validated in industrial & real-world scenarios Ready for integration into multiple robotic platforms |

Figure 2. Proposed soft-rigid hand layout, sensing, actuation, embedded electronics, applications and modular integration.

Closed-Loop System Architecture

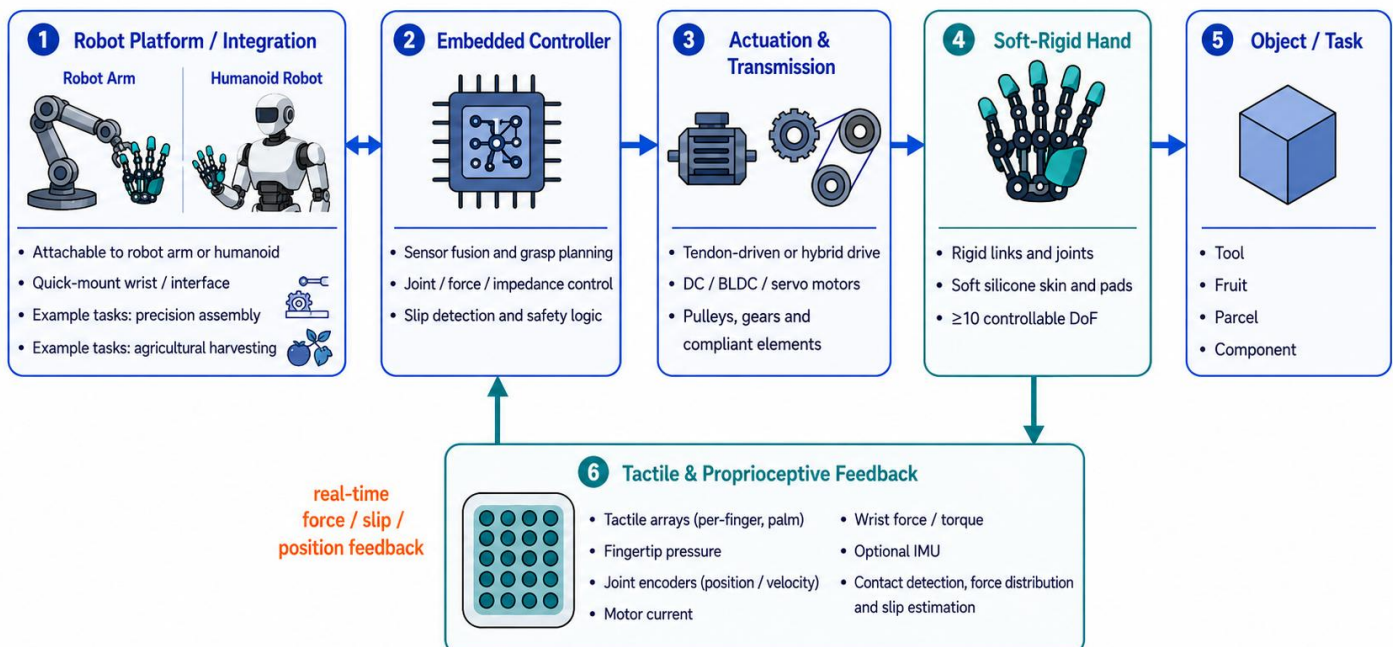


Figure 3. Block diagram of the tactile perception, embedded control, actuation and feedback architecture.

4. Suggested Hardware and Software Subsystems

Mechanical / Actuation	Sensing / Electronics	Control / Integration
• Rigid lightweight finger links and palm • Soft silicone skin and replaceable fingertip pads • Tendon-driven, direct-drive or hybrid actuators • Pulleys/gears, compliant transmission and quick wrist mount	• Joint encoders or magnetic position sensors • Tactile arrays and fingertip pressure sensors • Motor-current and wrist force/torque sensing • MCU/embedded computer, drivers, conditioning and power management	• ROS 2 host interface and robot-arm integration • Joint, force, impedance and slip-correction control • Object detection, pose estimation and grasp planning • CAN/EtherCAT/UART/USB/Ethernet communication and data logging

5. Closed-Loop Grasping and Intelligent Manipulation

Every grasp will operate as a feedback process: the system identifies a suitable grasp, the fingers approach the object, tactile sensors confirm contact, force and slip are estimated, and the controller adjusts position or tendon tension until the object is stable. The controller must generate a firm grasp for tools and heavy components while remaining sufficiently compliant for fruit, thin plastic parts and human–robot handovers.

1 Detect & plan	2 Approach	3 Touch	4 Sense force / slip	5 Adjust grip	6 Hold / manipulate
-----------------	------------	---------	----------------------	---------------	---------------------

Continuous tactile feedback returns from steps 3–6 to the embedded controller for millisecond-level correction.

Control and Perception Functions	Representative Application Demonstrations
• Joint position and velocity control with tendon-tension or motor-current regulation. • Fingertip force distribution, impedance/compliance control and safety limits. • Contact detection, slip estimation and automatic grip-force correction. • RGB/RGB-D object recognition, pose estimation and grasp-point selection. • Visual–tactile sensor fusion and preset power, pinch, tripod, lateral and precision grasps.	• Industry: precision assembly, tool operation and rapid switching between product types. • Agriculture: delicate harvesting of berries, tomatoes or apples with reduced bruising. • Logistics: grasping boxes, bottles, bags and irregular mixed-shape parcels. • Humanoid and service robotics: human–robot handover, assistance and everyday object use. • Modular integration: installation on robot arms, cobots, mobile manipulators and humanoids.

6. Implementation, Testing and Evaluation

PROJECT DEVELOPMENT, VALIDATION AND PERFORMANCE EVALUATION

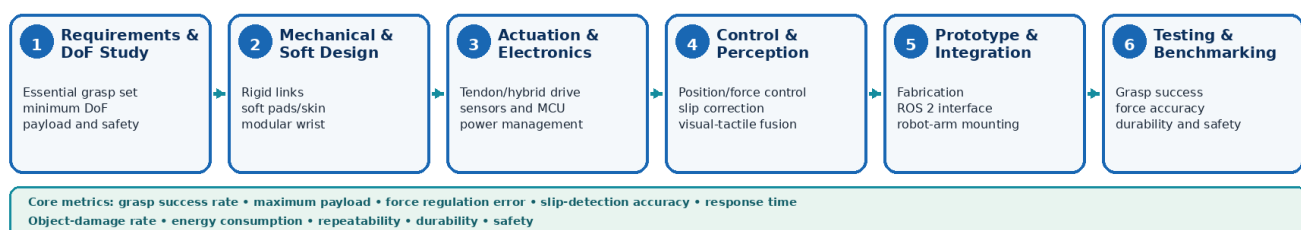


Figure 4. Proposed development workflow and quantitative evaluation plan.

Key Performance Measures	Expected Project Outcomes
• Grasp success and repeatability • Maximum payload and fingertip force • Force-control error and response time • Tactile/slip-detection accuracy • Energy, weight, durability and object damage	• Working modular soft–rigid hand prototype • Integrated tactile, force and joint sensing • Embedded real-time control and ROS 2 interface • Validated grasp library and application demos • Design files, software, test dataset and final report

For more details contact:

Prof. Mohammed Ibrahim mohammed.awad@eng.asu.edu.eg